

Closing the Algebra Is Not Enough: Lie-Closure Regularization Makes Learned Transformation Structure Comparable, Not Identifiable

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Abstract

Representations learned by different training runs are usually compared through pointwise similarity of their embeddings. We ask a structural question instead: do independently trained models learn the same *transformation algebra*? We study BracketNet, a transition-based representation learner that penalizes the component of each commutator of its learned skew-symmetric generators that leaves their span, trained with a fit–close–refit schedule. Our benchmark embeds $\text{SO}(2)$, T^2 and $\text{SO}(3)$ actions in a larger $\mathfrak{so}(d)$ and observes them through a nonlinear map. We froze the protocol on development seeds and evaluated on 20 fresh seeds; three findings follow.

(1) Long training on transition losses alone recovers the abelian T^2 algebra in most runs, yet leaves the $\text{SO}(3)$ generator span non-closed in 15/20 runs. (2) Closure regularization lowers $\text{SO}(3)$ closure on 20/20 seeds and roughly halves the distance to the ground-truth algebra ($0.4754 \rightarrow 0.2429$). It also makes the algebras of independently trained models more similar on disjoint seed pairs (Holm-corrected $p = 8.8 \times 10^{-3}$), with no detectable change in transport error. (3) Closure is not recovery. 8/20 $\text{SO}(3)$ runs close onto a chiral $\mathfrak{su}(2)$, one of the incorrect conjugacy classes we characterize, so cross-model agreement improves largely because disagreement becomes *discrete*.

Whether transformations transfer between two models tracks whether both recovered the correct algebra, irrespective of training objective. Linear CKA between latent spaces differs by at most 0.005 across methods, so it is essentially insensitive to these structural differences. We give corrected invariance statements, a rigorous bound on how far products of near-identity elements leave the learned span, and the classification underlying our failure analysis.

1 Introduction

When two networks are trained on the same data, do they learn the same thing? Representation-similarity measures answer this for *embeddings*: CKA, Procrustes and shape metrics compare point clouds up to a class of transformations [12, 18]. For models that learn how data *transform*, a second question matters: does the learned transformation structure agree? Here that structure is a set of latent generators whose exponentials act on the representation. Two models can embed data similarly yet represent transformations through incompatible algebras, in which case actions inferred by one model are meaningless in the other.

Transition-based symmetry learners [11, 14, 20] fit local transformations, but nothing in a transition loss requires the learned generators $A_1, \dots, A_K$ to span a Lie algebra, i.e. $[A_i, A_j] \in \text{span}\{A_k\}$. Without closure, the nominal K -parameter family is not locally closed under composition, and its “structure constants” are not defined. Closure is therefore a natural candidate for a *structural* inductive bias: it restricts each model to a subalgebra of $\mathfrak{so}(d)$. In the low-dimensional settings we study, these

fall into a few conjugacy classes (Proposition 4), which can be compared across models; in general the classification is richer, e.g. one-dimensional subalgebras of a torus form a continuum.

We test this idea carefully and report what holds and what does not. We fixed every choice on development seeds, with selection rules committed before the runs, and evaluated once on fresh seeds with a pre-registered primary hypothesis family. Our contributions:

- **A precise account of what a closure loss can guarantee** (Sec. 4). We give its exact behaviour under changes of generator coordinates (invariant only under orthogonal changes; its zero set is $GL(K)$ -invariant), and a rigorous bound showing that how far products of near-identity elements leave the learned span vanishes with the closure defect. We also classify the closed subalgebras in our benchmarks: for $SO(3)$ inside $\mathfrak{so}(4)$, closure admits two incorrect, chiral classes.
- **A structural evaluation** that separates closure, ground-truth recovery, cross-model agreement and action accuracy, together with **cross-model transformation transfer**: actions inferred by one model are applied in another model’s aligned latent space.
- **Evidence on 20 fresh seeds** (Secs. 6–8). Closure regularization substantially improves algebraic consistency for $SO(3)$ at no detectable transport cost, but turns continuous disagreement between models into discrete disagreement among closed classes, including frequent wrong ones. Transfer quality tracks structural correctness, not the training objective.

2 Related work

Learning transformation groups. Lie-group transformation models have a long history in unsupervised learning [4, 15, 17]. Recent methods discover symmetries from predictors [6], adversarially from distributional invariance [19], in learned latent spaces [20], beyond affine families [16], or as one-parameter transformation families [7]. Equivariant architectures instead assume the group is known [3, 5].

Group-structured representations from transitions. Symmetric embedding networks [14] and Homomorphism Autoencoders [11] learn latent spaces in which observed transitions act through a group representation. Symmetry-based disentanglement [9] defines representations through group actions. We share this setting. Our question is different: *whether the learned infinitesimal structure is a Lie algebra at all, and whether independently trained models agree on it.*

Comparing and aligning representations. CKA [12], Procrustes-type shape metrics [18] and relative representations [13] compare or align latent spaces of different models up to a class of transformations. The convergence of learned representations across models is an active hypothesis [10]. These tools compare *point clouds* of embeddings. We add a comparison of the *transformation structure* acting on them: after aligning two latent spaces, do their generator spans coincide, and do actions inferred by one model act correctly in the other?

3 Problem setting and method

Setting. Observations $x = h(u) \in \mathbb{R}^p$ arise from states u through an unknown injective map h . We observe unlabeled triples $(x_0, x_1, x_2) = (h(u), h(g_1 u), h(g_2 g_1 u))$, where g_1, g_2 are near-identity elements of an unknown connected compact group. The generator count K is given.

We learn an encoder $E : \mathbb{R}^p \rightarrow \mathbb{R}^d$, a decoder D , an action-inference network $a = q(z, z') \in \mathbb{R}^K$ (bounded as $0.65 \tanh(\cdot)$), and generators $A_1, \dots, A_K \in \mathfrak{so}(d)$. A coefficient vector acts as $\rho(a) = \exp(\sum_k a_k A_k) \in SO(d)$. Each A_k is antisymmetrized and rescaled to $\|A_k\|_F = \sqrt{2}$ at every forward pass.

Objectives. With $z_t = E(x_t)$ and $a_{st} = q(z_s, z_t)$:

- reconstruction $\mathcal{L}_{\text{rec}} = \sum_t \|D(z_t) - x_t\|^2$;
- one-step transport in latent and observation space, $\mathcal{L}_{\text{loc, lat}} = \sum_{t < 2} \|\rho(a_{t,t+1})z_t - z_{t+1}\|^2$ and $\mathcal{L}_{\text{loc, obs}} = \sum_{t < 2} \|D(\rho(a_{t,t+1})z_t) - x_{t+1}\|^2$;
- composition $\mathcal{L}_{\text{comp}} = \|\rho(a_{02}) - \rho(a_{12})\rho(a_{01})\|_F^2$;
- the Lie-closure residual

$$\mathcal{L}_{\text{close}} = \frac{1}{K^2} \sum_{i,j=1}^K \|(I - P_\varepsilon) \text{vec}[A_i, A_j]\|_2^2, \quad P_\varepsilon = B(B^\top B + \varepsilon I)^{-1} B^\top, \quad (1)$$

where B stacks the vectorized generators and $\varepsilon = 10^{-6}$;

- the Gram penalty $\|B^\top B - 2I\|_F^2$ and the anti-collapse term $\|\text{Cov}(z) - I\|_F^2$.

The LOCAL baseline minimizes $4\mathcal{L}_{\text{rec}} + 32\mathcal{L}_{\text{loc, lat}} + 2\mathcal{L}_{\text{loc, obs}}$ plus the two regularizers. +COMP. adds $1.2\mathcal{L}_{\text{comp}}$. BRACKETNET adds $\lambda(t)\mathcal{L}_{\text{close}}$ with a *fit-close-refit* schedule: $\lambda(t) = 0$ for the first 35% of updates, a linear ramp to λ by 70%, and frozen generators for the last 30% while E, D, q adapt. All three use identical networks, data and optimizer.

4 Theoretical analysis: what closure does and does not buy

Write $\mathfrak{a} = \text{span}(A)$, $G = B^\top B$, P_0 for the orthogonal projector onto $\mathfrak{a}$ and $N = I - P_0$. All statements below are proved in Appendix A and checked numerically by unit tests in the supplementary code.

Invariances of the objective. With $\varepsilon = 0$, a change of generator coordinates $B \mapsto BM$, $M \in GL(K)$, gives $\mathcal{L}_{\text{close}}(BM) = K^{-2} \|R(M \otimes M)\|_F^2$, where R is the residual matrix. Hence $\sigma_{\min}(M)^4 \mathcal{L}_{\text{close}} \leq \mathcal{L}_{\text{close}}(BM) \leq \sigma_{\max}(M)^4 \mathcal{L}_{\text{close}}$.

The loss is therefore invariant exactly under orthogonal M , it is not invariant under general invertible M , and its zero set, “ $\mathfrak{a}$ is a subalgebra”, is $GL(K)$ -invariant. For $\varepsilon > 0$ the regularized projector satisfies $P_\varepsilon(BM) = B(G + \varepsilon M^{-\top} M^{-1})^{-1} B^\top$, so it is not even a function of the span. A genuinely span-only objective evaluates $\mathcal{L}_{\text{close}}$ on the orthonormal basis $\sqrt{2}BG^{-1/2}$; we test it as a separate variant (Sec. 8).

Because each generator is normalized to $\sqrt{2}$, $\text{diag } G = 2$. The Gram penalty therefore acts only on off-diagonal entries, and its target $(2I \text{ or } I)$ does not affect gradients.

Proposition 1 (Exact closure integrates). *If $A_1, \dots, A_K \in \mathfrak{so}(d)$ are linearly independent and $[A_i, A_j] \in \mathfrak{a}$ for all i, j , then $\mathfrak{a}$ is a K -dimensional Lie subalgebra. It is the Lie algebra of a unique connected, immersed (not necessarily closed) subgroup $H \subseteq SO(d)$ generated by $\exp(\mathfrak{a})$.*

Let $\delta(\mathfrak{a}) = \sup_{X, Y \in \mathfrak{a} \setminus 0} \|N[X, Y]\|_F / (\|X\|_F \|Y\|_F)$ be the uniform escape constant.

Lemma 1 (The loss controls uniform escape). *$\delta(\mathfrak{a}) \leq K \sqrt{\mathcal{L}_{\text{close}}(A; 0)} / \lambda_{\min}(G)$. For an orthogonal basis normalized to $\sqrt{2}$, this is $\frac{K}{2} \sqrt{\mathcal{L}_{\text{close}}}$.*

Proposition 2 (Approximate closure controls BCH escape). *For $X, Y \in \mathfrak{a}$ with $\|X\|_F, \|Y\|_F \leq r < \frac{\ln 2}{4}$,*

$$\text{dist}(\log(e^X e^Y), \mathfrak{a}) \leq \frac{\delta}{4} [-\ln(2 - e^{4r}) - 4r] \leq \frac{1}{2} \delta r^2 + C(r_0) \delta r^3 \quad (r \leq r_0),$$

where $C(r_0) = (\frac{1}{4}[-\ln(2 - e^{4r_0}) - 4r_0] - 4r_0^2) / r_0^3$.

Both bounds vanish under exact closure, consistent with Proposition 1. The radius is small: it covers rotations by at most about 0.12 rad. The bound is therefore a local guarantee about the learned family near the identity, not about every action the coefficient network can output.

Proposition 3 (Chained orthogonal stability). *If $z_{t+1} = R_t z_t + e_t$ and $\hat{z}_{t+1} = R_t \hat{z}_t$ with $R_t \in SO(d)$, then $\|\hat{z}_T - z_T\| \leq \|\hat{z}_0 - z_0\| + \sum_{t < T} \|e_t\|$.*

Closure is not recovery. We separate four properties, each implied by the next: (i) closure; (ii) the correct conjugacy class of subalgebra of $\mathfrak{so}(d)$; (iii) ground-truth recovery in the learned coordinates, i.e. $\text{span}(RaR^\top) = \mathfrak{g}$ after the latent alignment R determined by the data; (iv) additionally, correct action inference.

Proposition 4 (Closed subalgebras in the benchmarks). *(a) Every non-degenerate 2-dimensional subalgebra of $\mathfrak{so}(5)$ is abelian and conjugate to the true $\mathfrak{t}^2$. Hence, for T^2 , (i) $\Rightarrow$ (ii). (b) Every 3-dimensional subalgebra of $\mathfrak{so}(4) \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$ is conjugate either to the diagonal $\mathfrak{so}(3)$ acting on $\mathbb{R}^3 \oplus \mathbb{R}$ (the true action) or to a chiral factor $\mathfrak{su}(2)_L$ or $\mathfrak{su}(2)_R$. These factors are exchanged by a reflection and are not conjugate to the diagonal $\mathfrak{so}(3)$. Hence, for $SO(3)$, (i) $\nRightarrow$ (ii). (c) No closed 4-dimensional subspace of $\mathfrak{so}(4)$ contains the diagonal $\mathfrak{so}(3)$. Hence, with $K=4$ for $SO(3)$, exact closure excludes the true algebra.*

The two $SO(3)$ classes have identical abstract structure but differ in representation. The conjugation-invariant participation ratio $\text{tr}(X^\top X)^2 / \text{tr}((X^\top X)^2)$ of unit elements is 2 for the diagonal class and 4 for a chiral factor. The mean commutator norm differs as well ($\sqrt{2}$ vs. 2 under our normalization), so a commutator threshold identifies only abelian versus non-abelian algebras, not the representation. No step of the objective forces (iii) or (iv). We claim no identifiability result.

5 Experimental protocol

Benchmarks. The groups are $SO(2)$, the abelian torus T^2 and the non-abelian $SO(3)$. Each acts linearly on a latent state in $\mathbb{R}^d$ that includes an extra invariant coordinate, with $d = 3, 5, 4$. The true algebra is therefore a proper subspace of a larger $\mathfrak{so}(d)$, and a random span does not close: the mean closure of random normalized spans is 0.60 for $K=2$ in $\mathfrak{so}(5)$ and 0.66 for $K=3$ in $\mathfrak{so}(4)$.

Initial states are drawn as $z_0 \sim \mathcal{N}(0, I_d)$, and transitions apply $\exp(\sum_k a_k T_k)$ with $a_k \sim \mathcal{N}(0, 0.24^2)$. Observations are $x = Qz + 0.18 \tanh(Bz)$ with Q orthogonal and B fixed; there is one fixed map per group, shared by all splits and seeds, and it is injective by construction. Each run trains on 1,400 triples and is evaluated on 500 ten-edge sequences generated from its own seed.

Development-only selection, then a frozen protocol. The training budget and closure weight are not fixed in advance; we chose them on five development seeds (5–9) with rules committed before any development run (Appendix B).

- *Budget:* the smallest budget at which the +COMP. baseline’s development transport is within 10% of its value at 4,000 updates. This rule is method-agnostic.
- *Closure weight:* the λ with the lowest development closure whose development transport is no worse than +COMP.’s.
- *Closure objective:* a span-only variant would replace Eq. (1) only if it were better on both criteria.

The rules selected 4000 updates, $\lambda = 3$ and Eq. (1). All other settings were fixed before selection; the full frozen protocol is in the supplement. We also evaluate $\lambda = 0.1$ at the same budget as a reference: it is the weight originally planned for this method. The final iterate is evaluated, with no early stopping.

Fresh evaluation. All methods are trained on 20 previously unused seeds (100–119). Ablations use a further 10 seeds (200–209). The structural categories and all thresholds were fixed before these runs:

- *collapsed:* $\sigma_{\min} / \sigma_{\max}(B) < 0.1$;
- *closed:* closure at most 1% of the random-span level;

Table 1: Fresh-seed results (seeds 100–119, mean $\pm$ s.d.; frozen protocol: 4000 updates, $\lambda = 3$). GT dist. is the ground-truth algebra distance after Procrustes alignment (chance: 0.80 for T^2 , 0.50 for SO(3)). CC is the number of correct-closed runs. Closure is identically 0 for the one-generator SO(2).

Group	Method	1-step MSE	10-step MSE	Closure residual	GT dist.	CC
SO(2)	Local	0.0026 $\pm$ 0.0010	0.0061 $\pm$ 0.0019	0 $\pm$ 0	0.000 $\pm$ 0.000	20/20
SO(2)	+Comp.	0.0026 $\pm$ 0.0010	0.0065 $\pm$ 0.0028	0 $\pm$ 0	0.000 $\pm$ 0.000	20/20
SO(2)	BracketNet	0.0026 $\pm$ 0.0010	0.0063 $\pm$ 0.0027	0 $\pm$ 0	0.000 $\pm$ 0.000	20/20
SO(2)	BracketNet ($\lambda=0.1$)	0.0026 $\pm$ 0.0010	0.0063 $\pm$ 0.0027	0 $\pm$ 0	0.000 $\pm$ 0.000	20/20
T^2	Local	0.0080 $\pm$ 0.0076	0.0645 $\pm$ 0.0841	$3.4 \times 10^{-1} \pm 7.2 \times 10^{-1}$	0.161 $\pm$ 0.282	14/20
T^2	+Comp.	0.0081 $\pm$ 0.0076	0.0605 $\pm$ 0.0757	$2.9 \times 10^{-1} \pm 6.3 \times 10^{-1}$	0.151 $\pm$ 0.274	14/20
T^2	BracketNet	0.0072 $\pm$ 0.0079	0.0539 $\pm$ 0.0792	$2.2 \times 10^{-2} \pm 5.6 \times 10^{-2}$	0.129 $\pm$ 0.282	16/20
T^2	BracketNet ($\lambda=0.1$)	0.0082 $\pm$ 0.0078	0.0622 $\pm$ 0.0785	$3.1 \times 10^{-1} \pm 6.1 \times 10^{-1}$	0.157 $\pm$ 0.277	14/20
SO(3)	Local	0.0048 $\pm$ 0.0027	0.0325 $\pm$ 0.0312	$6.3 \times 10^{-1} \pm 5.7 \times 10^{-1}$	0.479 $\pm$ 0.435	6/20
SO(3)	+Comp.	0.0048 $\pm$ 0.0031	0.0328 $\pm$ 0.0299	$6.2 \times 10^{-1} \pm 5.7 \times 10^{-1}$	0.475 $\pm$ 0.433	5/20
SO(3)	BracketNet	0.0049 $\pm$ 0.0034	0.0319 $\pm$ 0.0341	$2.5 \times 10^{-2} \pm 7.1 \times 10^{-2}$	0.243 $\pm$ 0.254	8/20
SO(3)	BracketNet ($\lambda=0.1$)	0.0050 $\pm$ 0.0031	0.0484 $\pm$ 0.0484	$5.4 \times 10^{-1} \pm 5.4 \times 10^{-1}$	0.453 $\pm$ 0.413	5/20

Table 2: Pre-registered primary hypotheses, BracketNet vs. +COMP. (one-sided). Units are seeds ($n=20$) for H1–H2 and disjoint seed pairs ($n=10$) for H3. The interval is a two-sided 95% t -interval for the difference BracketNet–+COMP.; d_z is the paired effect size; “wins” counts units where BracketNet is lower. p is the exact sign-flip p -value; Holm-corrected over the six tests.

Hypothesis	Group	n	+Comp.	BracketNet	95% CI of diff.	d_z	wins	p	Holm p
H1 closure	T^2	20	0.2906	0.0216	[-0.5427, +0.0049]	-0.46	16/20	4.3e-05	2.1e-04
H2 ground-truth distance	T^2	20	0.1506	0.1289	[-0.0674, +0.0240]	-0.22	16/20	1.6e-01	3.1e-01
H3 cross-seed distance	T^2	10	0.2138	0.1981	[-0.1421, +0.1106]	-0.09	7/10	3.8e-01	3.8e-01
H1 closure	SO(3)	20	0.6163	0.0250	[-0.8550, -0.3278]	-1.05	20/20	9.5e-07	5.7e-06
H2 ground-truth distance	SO(3)	20	0.4754	0.2429	[-0.3348, -0.1301]	-1.06	19/20	1.2e-04	4.9e-04
H3 cross-seed distance	SO(3)	10	0.6845	0.3922	[-0.4424, -0.1423]	-1.39	9/10	2.9e-03	8.8e-03

- *correct-closed*: closed, with ground-truth algebra distance at most 10% of chance;
- *incorrect-closed*: closed but not correct.

The ground-truth algebra distance is the mean squared sine of the principal angles between $\text{span}(RAR^\top)$ and the true span, where R is the orthogonal Procrustes map from $E(x)$ to the true state.

Statistics. Seeds are the independent units. For cross-seed quantities we use the 10 *disjoint* seed pairs (100, 101), ..., (118, 119), the same for every method; we never treat the 190 overlapping pairs as independent. The pre-registered primary family compares BracketNet with +COMP., one-sided, on three quantities, each for T^2 and SO(3): closure (H1), ground-truth algebra distance (H2) and cross-seed algebra distance (H3). Each uses an exact paired sign-flip test with Holm correction over the six tests. Everything else is secondary and reported with two-sided 95% t -intervals, which we interpret only when they exclude zero.

6 Results on fresh seeds

Transition fitting closes the abelian algebra, but not SO(3). At 4000 updates the baselines already recover the true T^2 algebra in 14/20 (+Comp.) and 14/20 (Local) runs (Table 1). For SO(3), the generator span stays *non-closed* in 15/20 (+Comp.) and 14/20 (Local) runs, even though one-step and ten-step transport errors are low. Fitting local transitions well is therefore not sufficient for algebraic consistency of a non-abelian family. The same holds for BracketNet at the originally planned weight $\lambda=0.1$, which behaves like the baselines (15/20 non-closed).

Closure regularization improves algebraic fidelity for SO(3) without changing transport. With $\lambda = 3$, SO(3) closure falls from 0.6163 to 0.0250, lower on 20/20 seeds (Holm $p = 5.7 \times 10^{-6}$,

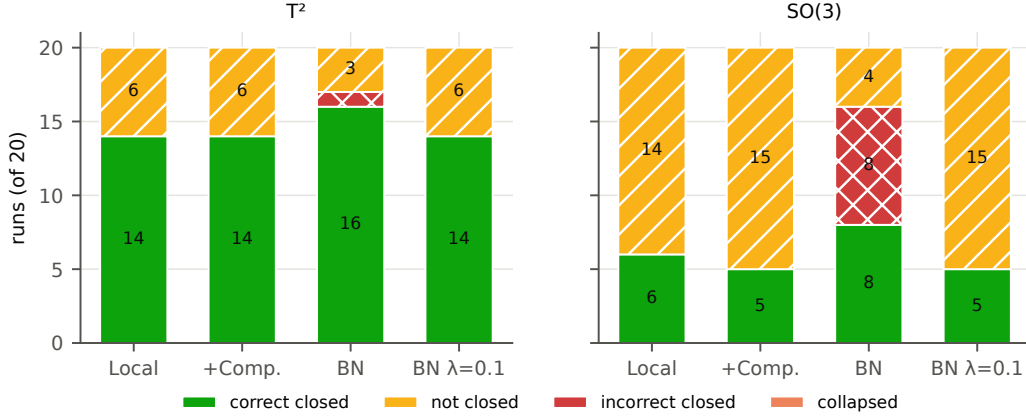


Figure 1: Structural category of each fresh-seed run (thresholds fixed in advance; BN = BracketNet). For $SO(3)$, closure turns non-closed runs into closed ones, about half of them chiral (incorrect).

Table 2). The ground-truth algebra distance falls from 0.4754 to 0.2429 (19/20 seeds, Holm $p = 4.9 \times 10^{-4}$).

For T^2 , closure is lower on 16/20 seeds (Holm $p = 2.1 \times 10^{-4}$), but the t -interval of the mean difference includes zero (upper end +0.0049): the baseline’s mean is driven by a few non-closed runs. The improvement in T^2 ground-truth distance is not significant (Holm $p = 0.31$).

Ten-step transport error does not change detectably in either group: the 95% intervals are $[-0.0199, 0.0066]$ for T^2 and $[-0.0080, 0.0061]$ for $SO(3)$.

But closure is not recovery: frequent chiral solutions. Figure 1 classifies every run. For $SO(3)$, BracketNet yields 8 correct-closed runs, 8 incorrect-closed runs (all 8 chiral, participation ratio ≈ 4) and 4 non-closed runs. +COMP. yields 5, 0 and 15 respectively.

The regularizer mostly converts non-closed runs into closed ones, and about half of these land in a wrong conjugacy class. This is exactly the possibility Proposition 4(b) leaves open. The correct-closed rate rises from 5/20 to 8/20, which is not significant (exact McNemar $p = 0.25$; 95% Wilson interval for BracketNet 22–61%). No run of any method collapsed.

7 Representation alignment: do independent models share an algebra?

Procedure. All models of a group share the observation map, so we can encode a common probe set (200 fresh sequences) with every model. For a seed pair (a, b) we fit the orthogonal Procrustes map R from $E_a(x)$ to $E_b(x)$ on that probe set, averaging both directions. We then measure four quantities:

- latent linear CKA;
- the *cross-seed algebra distance* between $\text{span}(RA^{(a)}R^\top)$ and $\text{span}(A^{(b)})$;
- the disagreement of inferred actions, $\|R\rho_a R^\top - \rho_b\|_F^2$, normalized by $\|\rho_b - I\|_F^2$;
- *transformation transfer*: on a disjoint held-out probe set, the action inferred by a is applied in b ’s latent space, $\hat{x}_{t+1} = D_b(R\rho_a(x_t, x_{t+1})R^\top E_b(x_t))$. We report the transfer gap $(\text{MSE}_{\text{transfer}} - \text{MSE}_{\text{native}})/(\text{MSE}_{\text{identity}} - \text{MSE}_{\text{native}})$, which is 0 when transfer is as good as b ’s own action and 1 when it is no better than doing nothing.

Units are the 10 disjoint seed pairs, identical for every method.

Table 3: Cross-seed alignment on 10 disjoint seed pairs (mean $\pm$ s.d.). Algebra distance: chance 0.80 (T^2), 0.50 (SO(3)). Transfer gap: 0 = native quality, 1 = identity guess. “Projected” re-expresses the transferred action in b ’s generator basis before exponentiating.

Group	Method	CKA	Algebra dist.	Action disagr.	Transfer gap	Projected gap
T^2	Local	0.920 $\pm$ 0.110	0.234 $\pm$ 0.261	0.596 $\pm$ 0.662	0.876 $\pm$ 1.621	0.647 $\pm$ 1.069
T^2	+Comp.	0.920 $\pm$ 0.109	0.214 $\pm$ 0.252	0.580 $\pm$ 0.659	0.852 $\pm$ 1.573	0.626 $\pm$ 1.032
T^2	BracketNet	0.925 $\pm$ 0.114	0.198 $\pm$ 0.341	0.485 $\pm$ 0.785	0.769 $\pm$ 1.406	0.432 $\pm$ 0.838
T^2	BracketNet ($\lambda=0.1$)	0.923 $\pm$ 0.106	0.228 $\pm$ 0.259	0.592 $\pm$ 0.662	0.770 $\pm$ 1.398	0.611 $\pm$ 0.997
SO(3)	Local	0.951 $\pm$ 0.074	0.685 $\pm$ 0.312	2.280 $\pm$ 0.995	0.745 $\pm$ 1.033	0.923 $\pm$ 0.413
SO(3)	+Comp.	0.949 $\pm$ 0.073	0.685 $\pm$ 0.308	2.222 $\pm$ 0.973	0.741 $\pm$ 1.008	0.909 $\pm$ 0.393
SO(3)	BracketNet	0.952 $\pm$ 0.073	0.392 $\pm$ 0.257	2.015 $\pm$ 0.815	0.732 $\pm$ 0.919	1.257 $\pm$ 0.536
SO(3)	BracketNet ($\lambda=0.1$)	0.949 $\pm$ 0.074	0.664 $\pm$ 0.296	2.194 $\pm$ 0.949	0.726 $\pm$ 0.978	0.924 $\pm$ 0.375

Algebras agree more across seeds, and CKA does not see it. For SO(3), the cross-seed algebra distance falls from 0.6845 (+Comp.) to 0.3922 (BracketNet), lower in 9/10 pairs (Holm $p = 8.8 \times 10^{-3}$; Table 2). For T^2 the difference is not significant (Holm $p = 0.38$). Over the same pairs, latent CKA is 0.949 vs. 0.952 for SO(3) and 0.920 vs. 0.925 for T^2 (Table 3). Pointwise similarity of embeddings carries essentially no information about whether two models represent transformations by the same algebra.

Why agreement improves: disagreement becomes discrete. Classifying each BracketNet SO(3) pair by the categories of its two models (Fig. 2) gives:

- 1 pair with both models correct;
- 1 pair with both models in the *same* incorrect (chiral) class, at distance 0: agreement on a wrong algebra;
- 4 pairs that pair a correct model with a chiral one;
- 4 pairs involving a non-closed model.

The correct/chiral pairs sit at distance ≈ 0.5 . This value is fixed by the geometry: the diagonal $\mathfrak{so}(3)$ and a chiral factor meet at principal angles of 45° . For +COMP., all 10 pairs involve a non-closed model, and their distances spread over $[0, 1]$. The reduction in mean distance is therefore real, but it reflects convergence onto a small set of closed conjugacy classes, as Proposition 4 predicts, more than convergence onto the correct algebra. This is the sense in which closure makes learned structure *comparable* without making it *identifiable*.

Transfer quality tracks structural correctness. Mean transfer gaps barely differ between methods. For BracketNet minus +COMP., the 95% intervals are $[-0.289, 0.124]$ for T^2 and $[-0.161, 0.144]$ for SO(3). What matters is structure. Pooling the pairs of all methods (a descriptive analysis: the methods share seeds), T^2 pairs in which both models are correct-closed transfer almost perfectly: median gap 0.006 over 22 pairs. All other pairs have median gap 0.855 over 18 pairs. For SO(3) only 1 pair is correct on both sides (gap 0.032); the remaining 39 have median gap 0.487. Cross-model reuse of transformations is thus governed by whether both models recovered the right algebra, which no method here guarantees. The one near-exception is instructive: an SO(3) +Comp. pair that misses the pre-registered closure threshold, but whose algebras are within 0.01 of the truth, transfers well (gap 0.03).

8 Ablations and failure modes

Ablations use 10 further seeds (200–209), with the final configuration as reference on the same seeds (Appendix Table 6). The intervals below are 95% paired t -intervals of the difference from the reference; with $n = 10$, the smallest attainable exact p is 0.002.

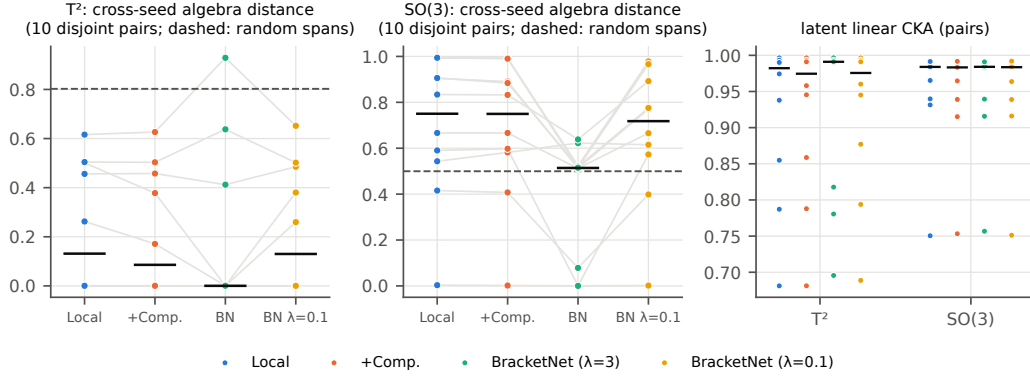


Figure 2: Cross-seed algebra distance on 10 disjoint seed pairs (BN = BracketNet); grey lines connect the same seed pair across methods. For $SO(3)$, BracketNet pairs concentrate at 0 (same class) and ≈ 0.5 (diagonal vs. chiral), while baseline pairs spread out. Right: latent CKA is nearly identical for all methods.

Replication. On these seeds, +COMP. again has higher $SO(3)$ closure ($+0.6571$, $[0.3016, 1.0126]$) and higher ground-truth distance ($+0.2176$, $[0.0786, 0.3567]$) than BracketNet. Correct-closed counts are 3/10 vs. 5/10.

The anti-collapse term is essential; the schedule is not demonstrably needed. Lowering the covariance weight from 16 to 1 destroys recovery: the T^2 ground-truth distance rises by 0.8755, and correct-closed runs fall to 0/10. The latent distorts (CKA with the true state 0.73 vs. 0.98), while the generators remain closed.

Replacing fit-close-refit by a constant penalty, or removing the generator freeze, changes neither $SO(3)$ ground-truth distance nor transport detectably:

- constant penalty: $[-0.1716, 0.0535]$ and $[-0.0234, 0.0394]$;
- no freeze: $[-0.0164, 0.0064]$ for ground-truth distance.

We therefore do not claim that staging is necessary.

Neither the Gram penalty nor the span-only closure objective makes a detectable difference. A three-times larger λ lowers closure slightly but leaves ground-truth distance and transport unchanged. A weaker λ ($0.3\times$) leaves more $SO(3)$ runs non-closed.

Misspecified generator count. With $K+1$ generators no run recovers the true algebra in either group. For $SO(3)$, closure regularization drives incorrect closed spans in most runs (0/10 correct). This is forced. A closed 4-dimensional subalgebra of $\mathfrak{so}(4)$ is of type $\mathfrak{su}(2) \oplus \mathfrak{u}(1)$, and the centralizer of the diagonal $\mathfrak{so}(3)$ in $\mathfrak{so}(4)$ is trivial, so *no* closed 4-dimensional span contains the true algebra. The learned spans capture on average 0.723 of it. With $K-1$, the learned algebra cannot contain the truth, and transport degrades sharply. Closure therefore presupposes the right K ; it cannot discover it.

Failure modes, summarized.

1. *Wrong closed class*: chiral $\mathfrak{su}(2)$ for $SO(3)$ (8/20 fresh runs). Transport cannot detect it: chiral runs fit transitions as well as correct ones.
2. *Seed-level fitting failures* shared by all methods (Sec. 9).
3. *Latent distortion* without the anti-collapse term.
4. *Misspecified K* .

Only (1) is introduced by the closure objective itself. Proposition 4 predicts it, and the participation-ratio diagnostic detects it without labels.

9 Limitations

Synthetic, small-scale benchmark. The groups are compact, low-dimensional and observed through a mild nonlinear map. We make no claim about images, physical trajectories or large pretrained models.

No identifiability. Closure did not prevent wrong closed algebras. Action inference is only as good as the recovered structure. Correct-closed $SO(3)$ runs have moderate action disagreement with the truth (0.36–0.43), while chiral and non-closed runs are no better than the identity (≈ 1 ; Appendix, Table 4).

Our latent alignment is orthogonal Procrustes, so a model whose encoder is far from an isometry of the true state space is scored as incorrect even if its algebra is conjugate to the truth (Proposition 4(a) guarantees such conjugacy for closed T^2 runs). The T^2 failures are of this kind. They are *seed-level* fitting failures: the same seeds fail under every method (102, 115 and 118 under all four, 109, 116 and 119 under all but the final BracketNet), all with high transport error.

Protocol choices. The budget and λ were selected on five development seeds. The λ rule needed an amendment, committed before any fresh run, because no candidate met its transport constraint; the selected $\lambda = 3$ trades some development transport for structure. The structural thresholds are conventions fixed in advance; different thresholds change the category counts but not the continuous metrics.

Scope of comparison. We compare controlled objective ablations under identical networks. We did not reimplement LieGAN, LaLiGAN or Homomorphism Autoencoders: they use different supervision signals (distributional invariance, or transitions with known group elements), so a fair comparison would require redesigning the benchmark. The theory is local: Proposition 2 covers only small group elements, and the generator count K is assumed known.

10 Conclusion

Explicit Lie-closure regularization makes the transformation structure learned by independent models more algebraically consistent and more comparable. For a non-abelian group it closes spans that transition fitting leaves open, roughly halves the distance to the true algebra, and reduces cross-seed disagreement, all without a detectable transport cost. It does not make that structure identifiable: closed but wrong (chiral) algebras are common, and agreement between models partly reflects agreement on a discrete set of classes, including wrong ones. Transferring transformations between models works when both recovered the correct structure, and largely fails otherwise. For representation alignment, the lesson is that comparing models requires comparing *what they do*, the algebra acting on the representation, and not only *where they embed*.

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A Proofs

Invariance law. Under $B \mapsto BM$, $[A'_i, A'_j] = \sum_{k,l} M_{ki} M_{lj} [A_k, A_l]$, so the commutator matrix becomes $C(M \otimes M)$, while $P_0(BM) = P_0(B)$. Hence $R \mapsto R(M \otimes M)$. The two-sided bound follows from $\sigma_{\min/\max}(M \otimes M) = \sigma_{\min/\max}(M)^2$. For $\varepsilon > 0$, $P_\varepsilon(BM) = BM(M^\top GM + \varepsilon I)^{-1} M^\top B^\top = B(G + \varepsilon M^{-\top} M^{-1})^{-1} B^\top$.

Proposition 1. The commutator is bilinear and antisymmetric and satisfies the Jacobi identity in every associative algebra, so closure makes $\mathfrak{a}$ a Lie subalgebra. The Lie subgroup theorem [8, Thm. 5.20] gives the unique connected Lie subgroup with Lie algebra $\mathfrak{a}$; it is generated by $\exp(\mathfrak{a})$. An irrational line in a torus shows that it need not be closed. $\square$

Lemma 1. Write $X = \sum_i x_i A_i$, $Y = \sum_j y_j A_j$ and $r_{ij} = N[A_i, A_j]$. By Cauchy–Schwarz over the K^2 index pairs, $\|N[X, Y]\| = \|\sum_i x_i y_j r_{ij}\| \leq \|x\| \|y\| (\sum \|r_{ij}\|^2)^{1/2} = \|x\| \|y\| K \sqrt{\mathcal{L}_{\text{close}}}$. Finally, $\|X\|^2 = x^\top G x \geq \lambda_{\min}(G) \|x\|^2$, and likewise for Y . $\square$

Proposition 2. Dynkin’s form of the BCH series [2] is $\log(e^X e^Y) = \sum_{m \geq 1} \frac{(-1)^{m-1}}{m} \sum \frac{[w]}{|w| \prod_i r_i! s_i!}$. The inner sum runs over words $w = X^{r_1} Y^{s_1} \dots X^{r_m} Y^{s_m}$ with $r_i + s_i > 0$; $[w]$ is the right-nested commutator and $|w| = \sum_i (r_i + s_i)$.

Since the Frobenius norm is submultiplicative, $\|[w]\| \leq 2^{|w|-1} r^{|w|}$. Let e_n bound $\|N[w]\|$ over words of length n ; then $e_1 = 0$ and $e_2 \leq \delta r^2$. Write $[w] = [W, Z]$ with $W \in \{X, Y\}$ and Z a nested commutator of length $n-1$. The term $[W, P_0 Z]$ has both arguments in $\mathfrak{a}$, so its escape is at most $\delta r \|Z\| \leq \delta 2^{n-2} r^n$, while $\|[W, NZ]\| \leq 2 r e_{n-1}$. By induction, $e_n \leq (n-1) 2^{n-2} \delta r^n$.

Each word therefore contributes at most $\frac{\delta}{4} (2r)^{|w|} / \prod_i r_i! s_i!$. Summing over all tuples gives $\sum_m \frac{1}{m} (e^{4r} - 1)^m = -\ln(2 - e^{4r})$ for $r < \ln 2/4$. Subtracting the length-1 words, which lie in $\mathfrak{a}$ and account for $4r$, gives the first bound.

The length-2 words sum exactly to $\frac{1}{2} [X, Y]$, whose escape is at most $\frac{1}{2} \delta r^2$; in the series bound they account for $4\delta r^2$. Replacing that amount gives the second bound. Since $-\ln(2 - e^u) - u$ has non-negative Taylor coefficients, the ratio $(\varphi(r) - 4r^2)/r^3$ is increasing, and $C(r_0)$ bounds it on $[0, r_0]$.

Finally, the series converges absolutely on this ball, and $\|e^X e^Y - I\| \leq e^{2r} - 1 < 1$ there. Both sides are analytic on the connected ball and agree near 0 [8, Thm. 5.3], so they agree on the whole ball. $\square$

Proposition 3. With $d_t = \hat{z}_t - z_t$ we have $d_{t+1} = R_t d_t - e_t$ and $\|R_t d_t\| = \|d_t\|$; iterate. $\square$

Proposition 4. Subalgebras of $\mathfrak{so}(d)$ are compact, hence reductive.

(a) A 2-dimensional reductive Lie algebra is abelian. Since $\text{rank } \mathfrak{so}(5) = 2$, it is a maximal abelian subalgebra, i.e. a Cartan subalgebra, and all Cartan subalgebras of a compact Lie algebra are conjugate [1].

(b) A 3-dimensional subalgebra $\mathfrak{h}$ of $\mathfrak{so}(4)$ is reductive and cannot be abelian, because the rank is 2. Hence $\mathfrak{h} \cong \mathfrak{su}(2)$. Its projections onto the two factors of $\mathfrak{so}(4) \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$ are each either zero or an isomorphism. If one projection is zero, $\mathfrak{h}$ is a factor. Otherwise $\mathfrak{h}$ is the graph of an automorphism, which is inner, so $\mathfrak{h}$ is conjugate to the diagonal subalgebra. The diagonal subalgebra acts on $\mathbb{R}^4 \cong \mathbb{C}^2 \otimes \mathbb{C}^2$ as $\mathbf{3} \oplus \mathbf{1}$, while a factor acts isoclinically. The participation ratio is a conjugation invariant that separates the two classes (2 vs. 4). A reflection in $O(4)$ exchanges the two factors.

(c) A 4-dimensional subalgebra is reductive of rank at most 2, so it is $\mathfrak{su}(2) \oplus \mathfrak{u}(1)$, with the $\mathfrak{su}(2)$ a factor (by the argument in (b)) and the $\mathfrak{u}(1)$ in the other factor. Its elements (X, Y) have Y confined to a line, whereas the diagonal $\mathfrak{so}(3)$ consists of all (X, X) . Equivalently, the centralizer of the diagonal $\mathfrak{so}(3)$ in $\mathfrak{so}(4)$ is trivial, so a fourth generator can never be added while preserving closure. $\square$

B Protocol details and provenance

Architecture and optimization. E , D and q are two-hidden-layer MLPs of width 64 with SiLU activations, with latent dimension d . Generators are initialized as $W \sim \mathcal{N}(0, 1/d)$. We use AdamW with learning rate 2×10^{-3} , weight decay 10^{-5} , batch size 160, global gradient-norm clipping at 5, $\epsilon = 10^{-6}$, Gram weight 1 and anti-collapse weight 16. The anti-collapse weight was chosen in a preliminary round on seeds 5–6 by a method-agnostic criterion: with weight 1, four of five latent covariance eigenvalues fell below 0.1.

Frozen generators receive no gradient, so they receive no weight decay either. All runs use one CPU thread. Repeated runs are bit-identical on the same machine.

Order of operations.

1. The selection rules were committed.
2. The development grid was run: budgets $\{420, 1000, 2000, 4000\} \times \{\text{LOCAL}, +\text{COMP.}, \text{BracketNet with } \lambda \in \{0.1, 3, 10, 30, 100\}\} \times \{T^2, \text{SO}(3)\} \times \text{seeds 5–9}$.
3. The rules were applied by a script, which wrote the frozen protocol file.
4. The closure-objective comparison was run on development seeds.
5. Seeds 100–119 and 200–209 were run once.

No hyperparameter was changed after any fresh-seed result was seen.

C Additional results

Table 4: Structural breakdown of fresh-seed runs (counts out of 20). Incorrect-closed runs are shown with the number of chiral ones in parentheses. “Bad action” counts correct-closed runs whose normalized action disagreement with the truth exceeds 0.5. PR is the participation ratio (diagonal $\text{so}(3)$: 2, chiral: 4; T^2 spans lie in $[2, 4]$).

Group	Method	Correct	Incorrect (chiral)	Non-closed	Collapsed	Bad action	Action disagr.	PR
T^2	Local	14	0 (0)	6	0	0	0.336 ± 0.438	2.98 ± 0.38
T^2	+Comp.	14	0 (0)	6	0	0	0.326 ± 0.429	2.93 ± 0.30
T^2	BracketNet	16	1 (0)	3	0	0	0.261 ± 0.405	2.86 ± 0.08
T^2	BracketNet ($\lambda=0.1$)	14	0 (0)	6	0	0	0.331 ± 0.433	2.93 ± 0.26
SO(3)	Local	6	0 (0)	14	0	0	0.849 ± 0.355	2.26 ± 0.44
SO(3)	+Comp.	5	0 (0)	15	0	0	0.838 ± 0.361	2.27 ± 0.44
SO(3)	BracketNet	8	8 (8)	4	0	0	0.764 ± 0.320	2.91 ± 1.00
SO(3)	BracketNet ($\lambda=0.1$)	5	0 (0)	15	0	0	0.833 ± 0.356	2.33 ± 0.46

Table 5: Development grid (seeds 5–9, used for selection only; means over 5 seeds). Each group of three columns gives 10-step MSE, closure residual and correct-closed count, for Local, +Comp. and BracketNet ($\lambda = 3$).

Group	Updates	MSE	Local closure	CC	MSE	+Comp. closure	CC	MSE	BracketNet closure	CC
T^2	420	0.1635	0.1374	1/5	0.1604	0.1104	1/5	0.1647	0.0600	1/5
T^2	1000	0.0704	0.0223	4/5	0.0678	0.0188	4/5	0.0676	0.0011	4/5
T^2	2000	0.0376	0.0016	4/5	0.0391	0.0022	4/5	0.0390	0.0005	4/5
T^2	4000	0.0308	0.0004	4/5	0.0312	0.0003	4/5	0.0317	0.0001	4/5
SO(3)	420	0.1391	0.8088	0/5	0.1532	0.7579	0/5	0.1479	0.3826	0/5
SO(3)	1000	0.0713	0.8762	1/5	0.0843	0.7771	1/5	0.0668	0.2032	1/5
SO(3)	2000	0.0396	0.9245	1/5	0.0308	0.6989	1/5	0.0573	0.1129	2/5
SO(3)	4000	0.0295	0.9574	1/5	0.0276	0.7086	1/5	0.0340	0.0568	2/5

Table 6: Ablations on seeds 200–209 (mean $\pm$ s.d. over 10 seeds). Coverage is the fraction of the true span contained in the learned span after alignment. The last column gives correct / incorrect / non-closed / collapsed counts. With a misspecified K , “correct” is impossible by definition.

Group	Variant	10-step MSE	Closure	GT dist.	Coverage	Categories
T^2	BracketNet (reference)	0.0157 $\pm$ 0.0047	$3.6 \times 10^{-5} \pm 2.0 \times 10^{-5}$	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000	10/0/0/0
T^2	+Comp. (reference)	0.0231 $\pm$ 0.0275	$2.0 \times 10^{-1} \pm 6.2 \times 10^{-1}$	0.050 $\pm$ 0.156	0.950 $\pm$ 0.156	9/0/1/0
T^2	constant penalty, no staging	0.0370 $\pm$ 0.0475	$1.2 \times 10^{-4} \pm 2.2 \times 10^{-4}$	0.076 $\pm$ 0.170	0.924 $\pm$ 0.170	8/2/0/0
T^2	staged ramp, no freeze	0.0156 $\pm$ 0.0048	$3.4 \times 10^{-5} \pm 2.1 \times 10^{-5}$	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000	10/0/0/0
T^2	no Gram penalty	0.0274 $\pm$ 0.0382	$1.4 \times 10^{-2} \pm 4.5 \times 10^{-2}$	0.027 $\pm$ 0.084	0.973 $\pm$ 0.084	9/0/1/0
T^2	anti-collapse weight 1	0.1997 $\pm$ 0.0253	$3.1 \times 10^{-3} \pm 4.6 \times 10^{-3}$	0.876 $\pm$ 0.130	0.124 $\pm$ 0.130	0/8/2/0
T^2	span-only (whitened) closure	0.0157 $\pm$ 0.0047	$3.6 \times 10^{-5} \pm 2.0 \times 10^{-5}$	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000	10/0/0/0
T^2	0.3 λ	0.0236 $\pm$ 0.0287	$1.4 \times 10^{-1} \pm 4.5 \times 10^{-1}$	0.040 $\pm$ 0.127	0.960 $\pm$ 0.127	9/0/1/0
T^2	3 λ	0.0156 $\pm$ 0.0044	$1.6 \times 10^{-5} \pm 1.1 \times 10^{-5}$	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000	10/0/0/0
T^2	$K+1$ generators	0.0248 $\pm$ 0.0194	$3.2 \times 10^{-1} \pm 1.1 \times 10^{-1}$	0.375 $\pm$ 0.103	0.937 $\pm$ 0.154	0/1/9/0
T^2	+Comp., $K+1$	0.0222 $\pm$ 0.0245	$8.2 \times 10^{-1} \pm 1.9 \times 10^{-1}$	0.367 $\pm$ 0.108	0.949 $\pm$ 0.161	0/0/10/0
T^2	$K-1$ generators	0.1917 $\pm$ 0.0300	0 $\pm$ 0	0.501 $\pm$ 0.001	0.499 $\pm$ 0.001	0/10/0/0
SO(3)	BracketNet (reference)	0.0218 $\pm$ 0.0160	$4.2 \times 10^{-2} \pm 8.0 \times 10^{-2}$	0.244 $\pm$ 0.298	0.756 $\pm$ 0.298	5/2/3/0
SO(3)	+Comp. (reference)	0.0393 $\pm$ 0.0309	$7.0 \times 10^{-1} \pm 5.2 \times 10^{-1}$	0.462 $\pm$ 0.412	0.538 $\pm$ 0.412	3/0/7/0
SO(3)	constant penalty, no staging	0.0298 $\pm$ 0.0394	$1.4 \times 10^{-2} \pm 2.6 \times 10^{-2}$	0.185 $\pm$ 0.298	0.815 $\pm$ 0.298	6/1/3/0
SO(3)	staged ramp, no freeze	0.0299 $\pm$ 0.0191	$2.6 \times 10^{-2} \pm 4.6 \times 10^{-2}$	0.239 $\pm$ 0.301	0.761 $\pm$ 0.301	5/2/3/0
SO(3)	no Gram penalty	0.0181 $\pm$ 0.0136	$1.5 \times 10^{-2} \pm 4.3 \times 10^{-2}$	0.177 $\pm$ 0.260	0.823 $\pm$ 0.260	6/2/2/0
SO(3)	anti-collapse weight 1	0.0302 $\pm$ 0.0140	$5.6 \times 10^{-2} \pm 2.2 \times 10^{-2}$	0.619 $\pm$ 0.063	0.381 $\pm$ 0.063	0/1/9/0
SO(3)	span-only (whitened) closure	0.0182 $\pm$ 0.0138	$4.0 \times 10^{-2} \pm 7.7 \times 10^{-2}$	0.243 $\pm$ 0.298	0.757 $\pm$ 0.298	5/2/3/0
SO(3)	0.3 λ	0.0314 $\pm$ 0.0284	$2.2 \times 10^{-1} \pm 3.4 \times 10^{-1}$	0.299 $\pm$ 0.297	0.701 $\pm$ 0.297	4/1/5/0
SO(3)	3 λ	0.0274 $\pm$ 0.0244	$1.7 \times 10^{-2} \pm 3.0 \times 10^{-2}$	0.237 $\pm$ 0.302	0.763 $\pm$ 0.302	5/2/3/0
SO(3)	$K+1$ generators	0.0225 $\pm$ 0.0108	$1.2 \times 10^{-1} \pm 2.4 \times 10^{-1}$	0.457 $\pm$ 0.076	0.723 $\pm$ 0.102	0/7/3/0
SO(3)	+Comp., $K+1$	0.0156 $\pm$ 0.0087	$5.5 \times 10^{-1} \pm 2.6 \times 10^{-1}$	0.511 $\pm$ 0.220	0.652 $\pm$ 0.293	0/0/10/0
SO(3)	$K-1$ generators	0.1253 $\pm$ 0.0535	$6.0 \times 10^{-1} \pm 5.1 \times 10^{-1}$	0.464 $\pm$ 0.166	0.536 $\pm$ 0.166	0/4/6/0

D Reproducibility

The anonymized supplementary code contains everything needed to reproduce the paper:

- the data generator, the model and training code;
- the pre-registered selection rules and the generated frozen protocol;
- per-run JSON files for every development, fresh-seed and ablation run, and the fresh-seed checkpoints;
- the analysis script that produces every table and every number in the text (as $\text{\LaTeX}$ macros);
- the figure scripts and the mathematical unit tests.

Repeated runs are bit-identical on the same machine. All runs are CPU-only and take 31–106 s each for 4000 updates (four runs in parallel).

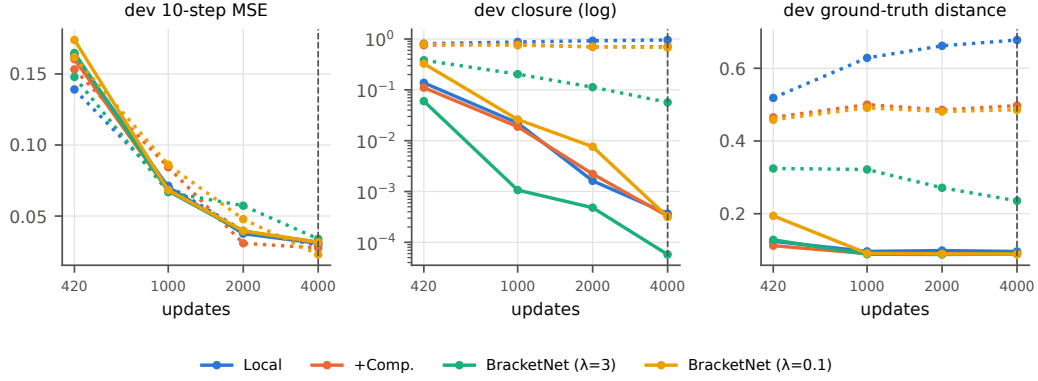


Figure 3: Development-seed budget sweep (solid: T^2 ; dotted: $SO(3)$). The vertical line marks the selected budget.

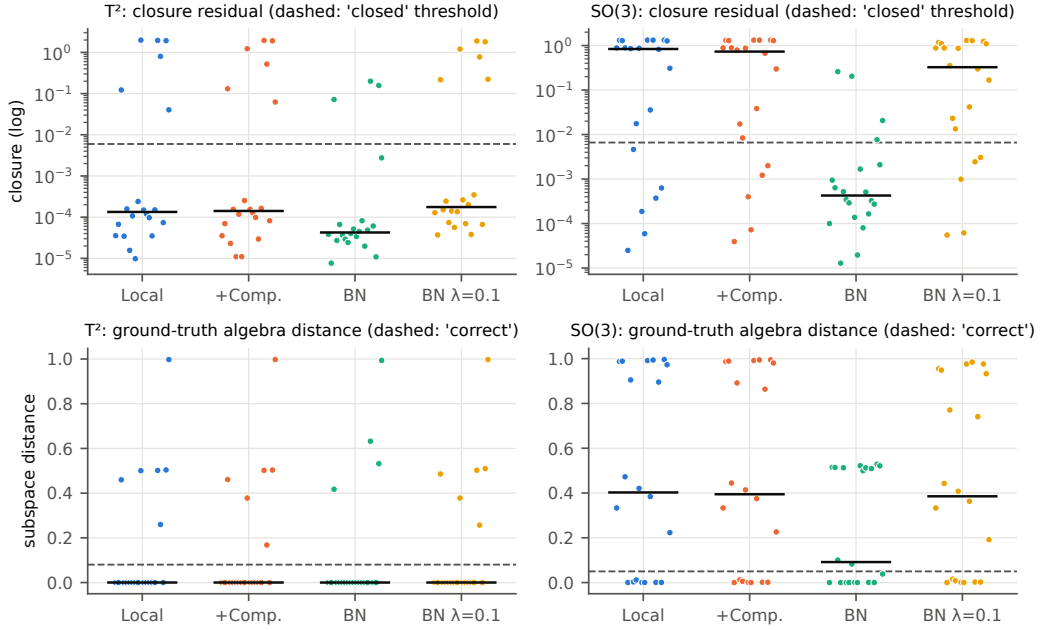


Figure 4: Per-seed closure residual (top) and ground-truth algebra distance (bottom) on fresh seeds; bars mark medians and dashed lines the pre-registered thresholds.

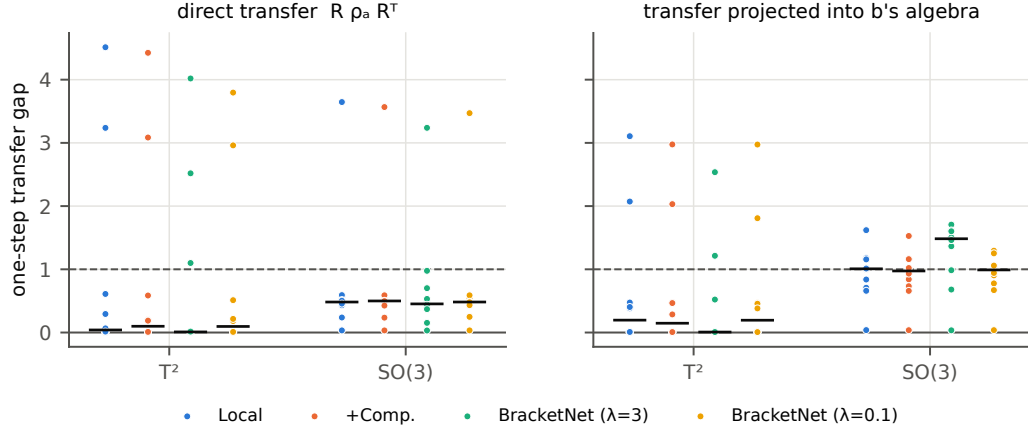


Figure 5: Cross-model transfer gap per disjoint seed pair.

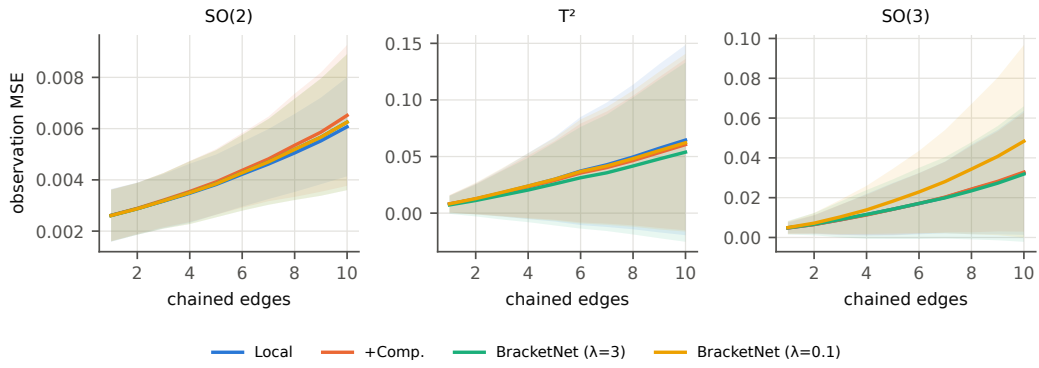


Figure 6: Chained transport error versus path length (mean $\pm$ s.d. over 20 fresh seeds).